

GRADUATION PROJECT PRESENTATION - MODULE

Daily Planners, Habits and Streaks Calculations

Structured logging schemas, habit streaks CTE queries, focus timers sync, and productivity metrics.

Presenter: Member 6

Core Focus: Productivity & Analytics

Menoufia University | Computer Science & Engineering Division

Daily Planner Logs & Checklist Schemas

- The daily planner coordinates checklists, mood trackers, and highlights.
- Daily task checklists reference parent daily logs.
- We index task lists by parent log IDs to optimize query speeds.
- [SCREENSHOT: User daily sheet showing mood ratings, highlights text, and finished task checkboxes]

```
-- Daily logs planner DDL
CREATE TABLE public.daily_logs (
  id UUID PRIMARY KEY,
  user_id UUID REFERENCES profiles(id),
  date DATE NOT NULL,
  rating TEXT,
  highlight TEXT
);
```

Mood Tracker Icon Mappings

- Students select mood icons to track study consistency.
- Icons are stored as emoji strings inside daily log tables.
- The app uses mood tags to filter productivity metrics.
- [SCREENSHOT: Mood selection panel screen view]

Mood Icon	Visual representation	Database Tag Value	Target Daily Rating
Happy Face	(Happy) Emoji	happy	Active study state
Neutral Face	(Neutral) Emoji	neutral	Normal study consistency
Sad Face	(Sad) Emoji	sad	Low energy study state
Flame Spark	(Fire) Emoji	fire	High focus consistency

Habits Tracker Scheduling Rules

- Habit schedules support Daily, Weekdays, Weekends, or Custom frequencies.
- The system checks date coordinates to filter active habits dynamically.
- This scheduling logic runs directly in the database engine.
- [SCREENSHOT: Habits tracker component displaying active tasks for weekdays]

```
-- Habit frequency check DDL
CREATE FUNCTION check_habit(habit_id UUID, check_date DATE)
RETURNS BOOLEAN AS $$
    -- Checks habit frequency rules...
    -- Returns true if active
$$ LANGUAGE sql;
```

Habit Streaks SQL Calculation (CTE)

- Cortex calculates habit streaks using a PostgreSQL CTE query.
- The query computes difference metrics between dates and row indexes.
- Grouping consecutive days allows the database to identify active streaks.
- [SCREENSHOT: User profile dashboard rendering habit streak counts and flame icons]

```
-- CTE Streak query snippet
WITH completions AS (
    SELECT date, ROW_NUMBER() OVER (ORDER BY date) as rn
    FROM habit_logs WHERE completed = TRUE
)
SELECT COUNT(*) as streak_length
FROM completions
GROUP BY (date - rn::int);
```

Pomodoro Focus Session Schema

- Focus sessions track study durations and completed task details.
- The table references courses, linking focus logs to subjects.
- We index sessions by user ID to optimize query performance.
- [SCREENSHOT: Web app focus page displaying active Pomodoro timer widgets]

```
-- Pomodoro focus session DDL
CREATE TABLE pomodoro_sessions (
  id UUID PRIMARY KEY,
  user_id UUID REFERENCES profiles(id),
  course_id UUID REFERENCES courses(id),
  duration_minutes INT,
  completed_at TIMESTAMPTZ
);
```

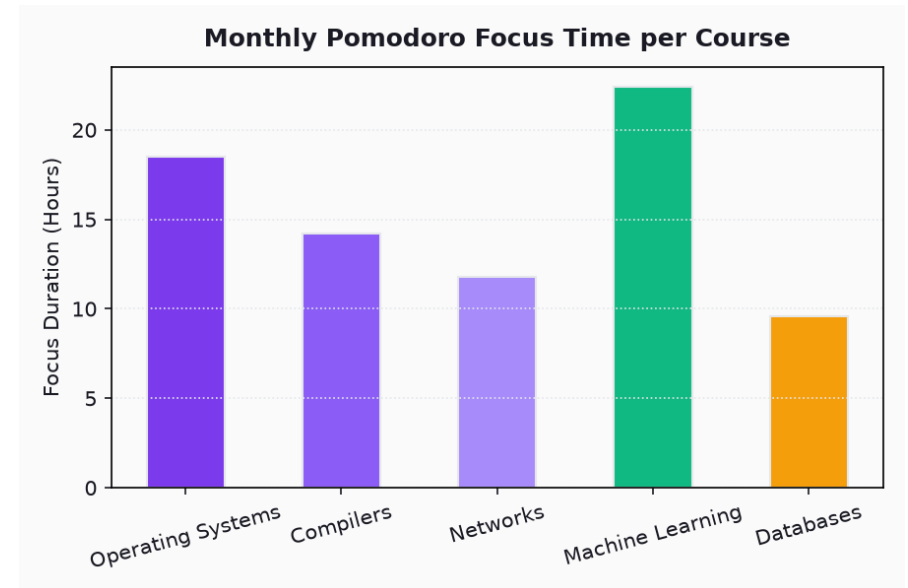
Focus Sessions Database View

- Cortex compiles study logs using database views.
- Views aggregate focus metrics to calculate total study hours per course.
- Pre-compiling these views reduces query sizes when loading reports.
- [SCREENSHOT: Productivity report displaying tables of focus durations and session charts]

```
-- Productivity metrics view
CREATE VIEW user_focus_summary AS
SELECT
    user_id,
    SUM(duration_minutes) / 60.0 as focus_hours
FROM pomodoro_sessions
GROUP BY user_id;
```

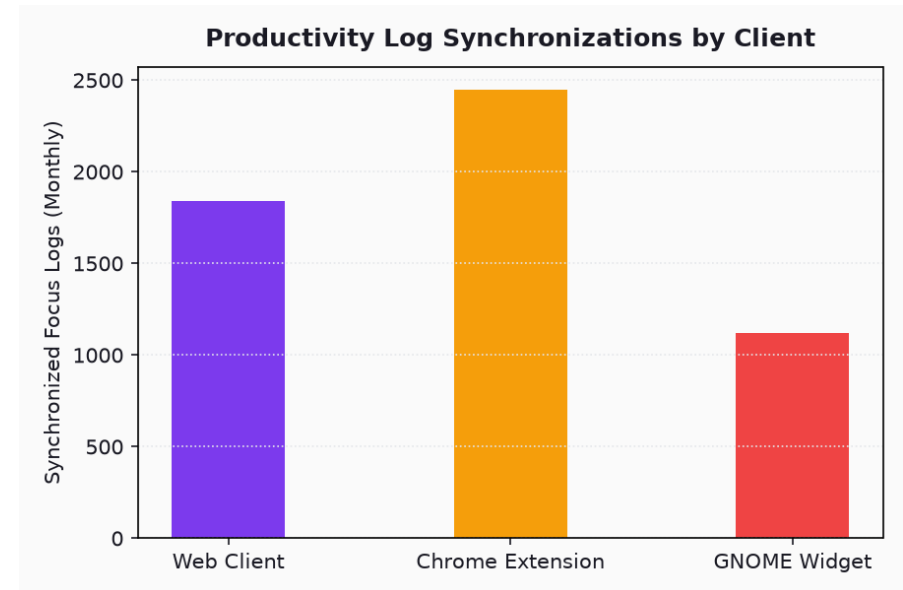
Pomodoro Focus Time per Course

- This bar chart shows focus hours logged across different courses.
- Subjects like Machine Learning show high focus times.
- These metrics help students evaluate their study habits over time.
- [SCREENSHOT: Productivity analytics dashboard showing focus hours charts per subject]



Multi-Client Focus Sessions Sync

- Cortex syncs focus logs from web, Chrome extension, and GNOME widget clients.
- Our browser extension and web clients log high focus activity.
- REST APIs coordinate this focus data, keeping metrics updated across devices.
- [SCREENSHOT: Synchronizer status log screen panels displaying client sync states]



Productivity Yearly Commit Heatmap

- Yearly grids display daily study contributions using contribution grids.
- Cell color intensity matches logged study hours.
- The canvas renders commit grids dynamically based on yearly metrics.
- [SCREENSHOT: GitHub-style commit grid displaying daily study contributions]

```
-- Daily activity query
SELECT
    date_trunc('day', completed_at) as study_day,
    COUNT(id) as sessions_count
FROM pomodoro_sessions
WHERE user_id = target_user_id
GROUP BY study_day;
```